FBISE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: 1HA-I 2309 — B(3091) / D(7091) / SP(3099)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

Note: CHEMISTRY HSSC-I. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 3091. Fill the relevant bubble against each question according to curriculum:

Q1. The volume occupied by 3.01×10^{23} molecules of NH_3 gas at STP is:

(A) $11.2 dm^3$ (B) $22.4 dm^3$

(C) $33.6 dm^3$ (D) $44.8 dm^3$

Q2. Lymen series of spectral lines lies in which of the following region of electromagnetic spectrum?

(A) UV-region (B) Near IR region

(C) Mid IR region (D) Visible region

Q3. Which of the following molecules does not possess trigonal planar geometry?

(A) PCl_3 (B) AlF_3

(C) NF_3 (D) NH_3

Q4. Which of the following aromatic compounds has zero dipole moment?

(A) Chloro Benzene (B) 1,2 – Dichloro Benzene

(C) 1,3 – Dichloro Benzene (D) Benzene

Q5. The value of general gas constant R is 62.4. Its units are:

(A) $dm^3.atmK^{-1}\ mol^{-1}$ (B) $cm^3.torrK^{-1}\ mol^{-1}$

(C) $dm^3\ mmHg\ K^{-1}\ mol^{-1}$ (D) $N.m\ K^{-1}\ mol^{-1}$

Q6. Rate of diffusion of H_2 gas (rH_2) as compared to that of He gas (rHe) is:

(A) $rH_2 = rHe$
(B) $rH_2 = 1.41rHe$
(C) $rH_2 = 4rHe$
(D) $rH_2 = 2rHe$

Q7. At 145°C, Cholestryl Benzoate exists as:

(A) Crystalline solid
(B) Non-Crystalline solid
(C) Liquid crystal
(D) Clear liquid

Q8. The increase in boiling point of hydrides of group IV-A (CH_4 to SnH_4) is due to increase in the strength of:

(A) Hydrogen bonding
(B) Dipole-dipole forces
(C) London dispersion forces
(D) Electrostatic forces

Q9. Which of the following is NOT an anisotropic property?

(A) Cleavage plane
(B) Electrical conductivity
(C) Co-efficient of thermal expansion
(D) Polymorphism

Q10. Identify the pair of substances which are not isomorphs?

(A) NaCl and KBr
(B) $ZnSO_4$ and $NiSO_4$
(C) $CaCO_3$ and $NaNO_3$
(D) Ag_2SO_4 and $CdSO_4$

Q11. The value of K_p will become equal to K_c, K_p and K_x when

(A) $\Delta n = 1$
(B) $\Delta n = -1$
(C) $\Delta n = 1/2$
(D) $\Delta n = 0$

Q12. pOH of 10^{-2}M solution of H_2SO_4 is:

(A) 2.0
(B) 14.0
(C) 1.7
(D) 12.3

Q13. For which of the following reaction, the rate constant is equal to the rate of reaction?

(A) Zero order reaction
(B) 1st order reaction
(C) 2nd order reaction
(D) 3rd order reaction

Q14. Boiling point of a solution prepared by dissolving 18g glucose in 1kg water is (K_b = 0.52):

(A) 100.52°C
(B) 100.052°C
(C) 101.52°C
(D) 101.052°C

Q15. A colloid in which a liquid is dispersed into another liquid is called:

(A) Sol
(B) Gel
(C) Aerosol
(D) Emulsion

Q16. Heat capacity of a substance is measured in the units of:

(A) JK^{-1} (B) $JK^{-1}g^{-1}$

(C) $JK^{-1}mol^{-1}$ (D) $Jmol^{-1}$

Q17. The number of electrons required to balance the equation are: $Cr_2O_7^{2-} + 14H^+ \rightarrow 2Cr^{3+} + 7H_2O$

(A) 2 (B) 3

(C) 5 (D) 6

Note: SUPPLEMENTARY TABLE (Atomic No / Symbol / Mass No): 1 H 1; 2 He 4; 3 Li 7; 4 Be 9; 5 B 11; 6 C 12; 7 N 14; 8 O 16; 9 F 19; 10 Ne 20; 11 Na 23; 12 Mg 24; 13 Al 27; 14 Si 28; 15 P 31; 16 S 32; 17 Cl 35.5; 18 Ar 40; 19 K 39; 20 Ca 40. —1HA-I 2309-3091—. Page 1 of 1.

Note: CHEMISTRY HSSC–I. Time allowed: 2:35 Hours. Total Marks Sections B and C: 68. NOTE: Answer any FOURTEEN parts from Section 'B' and attempt any TWO questions from Section 'C' on the separately provided answer book. Write your answers neatly and legibly.

Q. SECTION – B (Marks 42)

Q. 2 Answer any FOURTEEN parts from the following. All parts carry equal marks. (14 x 3 = 42)

(i) Calculate the mass of NH_3 gas produced when $8dm^3$ of H_2 gas is reacted with excess of N_2 gas. $N_2 + 3H_2 \rightarrow 2NH_3$

(ii) How much energy is required to remove an electron from 1st orbit of Li^{+2} ion in the units of J/atom and kJ/mol?

(iii) Write down the disadvantages of valence bond theory.

(iv) Describe any two factors which affect bond length. Give one example for each.

(v) Describe the geometries of following molecules on the basis of VSEPR theory. (a) SO_3 (b) PCl_3

(vi) What is an Isobar? Draw an isobar for a given mass of an ideal gas at 1 atm? How does the position of isobar change with increase in pressure?

(vii) How are London dispersion forces developed in a sample of Helium gas?

(viii) Which substance in each of the following pairs has stronger London dispersion forces? Give reasons: (a) Ar or Kr (b) Br_2 or I_2 (c) C_2H_6 or C_4H_{10}

(ix) What is the role of hydrogen bonding in: (a) Cleansing action of soap (b) Structure of DNA and protein molecules

(x) Describe transition temperature by giving two examples.

(xi) Differentiate between Hexagonal close packing and cubic close packing in the structure of metals.

(xii) What is a precipitation reaction? How can one predict the formation of precipitates of CaF_2 in the following reaction? $Ca^{+2} + 2F^- \rightleftharpoons CaF_2$

(xiii) How an Acetic acid/sodium acetate buffer resists its change in pH on addition of NaOH?

(xiv) Why vapour pressure of a solvent decreases when a non-volatile, non-electrolyte solute is dissolved in it?

(xv) What is reverse osmosis? Give its one application.

(xvi) If conc. of O_3 in the atmosphere reaches 0.5 ppb, what mass of O_3 would be present per kg of the air?

(xvii) Calculate $\Delta H°$ for the following reaction $2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$ the values of $\Delta H_f°$ for C_8H_{18}, O_2, CO_2 and H_2O are –269kJ, 0kJ, –393.5 kJ and –285kJ respectively.

(xviii) When 1.8g glucose is burnt in bomb calorimeter, the temperature of water increases for 25°C to 31.52°C. If heat capacity of calorimeter is 4.321kJ/K, calculate the heat of combustion of glucose.

(xix) Predict E°cell for Zn–Ni cell and write its cell reactions. The values of reduction potential of Zn and Ni are given as: E°Zn = –0.76 V, E°Ni = –0.25 V

(xx) Balance the equation by ion-electron method. $BrO_3^{-1} + SO_2 \rightarrow HSO_4^{-1} + Br^{-1}$

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. 100g Zn is reacted with 100g solution of HCl. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ (i) How much volume of H_2 gas is released at STP? (ii) Also determine the mass of non-limiting reactant in excess. b. Derive a relationship for energy reasled (ΔE) when an electron drops form n_2 orbit to n_1 orbit in He^+ ion. Also calculate ΔE if $n_1 = 1$ and $n_2 = 3$ in He^{+1} ion.

Q. 4 a. What is salt hydrolysis? Which type of cations and anions undergo hydrolysis? Describe the hydrolysis of four types of salts giving one example for each. b. What is collision theory? Describe with reference to the energy of activation, formation of activated complex and heat of reaction.

Q. 5 a. State and explain Dalton's law of partial pressure. Derive a relationship between: (i) Partial pressure and number of moles (ii) Partial pressure and mole fraction. b. Describe the effect of stated change on the following reactions at equilibrium position. (i) Decreasing the volume: $2NO + O_2 \rightleftharpoons 2NO_2$ (ii) Increasing temperature: $CH_4 + H_2O \rightleftharpoons CO + 3H_2$ $\Delta H = +256kJ$ (iii) Adding I_2: $2HI \rightleftharpoons H_2 + I_2$

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VERSION D

Note: CHEMISTRY HSSC-I. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 7091. Fill the relevant bubble against each question according to curriculum:

Q1. 22.4dm³ of which of the following gas contains greater number of atoms?

(A) Helium
(B) Chlorine
(C) Ozone
(D) Ammonia

Q2. The mass of O_2 required to burn 0.1 mole of C_2H_5OH as per following equation $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$

(A) 3.2g
(B) 32g
(C) 9.6g
(D) 96g

Q3. How much larger is the radius of 3rd orbit of Hydrogen atom as compared to the 1st one?

(A) 2 – times
(B) 3 – times
(C) 6 – times
(D) 9 – times

Q4. The values of quantum numbers for an electron present in 3d_xy orbital, are

(A) n = 3, l = 2, m = +2
(B) n = 3, l = 2, m = –2
(C) n = 3, l = 2, m = –1
(D) n = 3, l = 1, m = +1

Q5. The geometry of a molecule containing 2 – σ bond and one lone pair of electrons in the valence shell of central atom, is

(A) Trigonal bipyramid
(B) Trigonal pyramidal
(C) Trigonal planar
(D) Angular

Q6. Under what conditions, the deviation of a gas from ideal behavior, is maximum

(A) –10°C, 1atm
(B) –10°C, 5atm
(C) 0°C, 1atm
(D) 0°C, 5atm

Q7. Which of the following molecule has least polarizability?

(A) F_2
(B) Cl_2
(C) Br_2
(D) I_2

Q8. Graphite conducts the electricity in a direction parallel to the layers of carbon-atoms. This property is known as:

(A) Isotropy
(B) Anisotropy
(C) Allotropy
(D) Symmetry

Q9. Which of the following ionic compounds possesses greater lattice energy?

(A) LiCl
(B) NaCl
(C) KCl
(D) $MgCl_2$

Q10. For the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, K_p and K_c are related as:

(A) $K_p > K_c$
(B) $K_c > K_p$
(C) $K_c = K_p$
(D) $K_p = 1/2\ K_c$

Q11. The maximum yield of SO_3 is obtained in the reaction: $2SO_2 + O_2 \rightleftharpoons 2SO_3$ $\Delta H = -198kJ$ by

(A) Increasing temperature Increasing pressure
(B) Increasing temperature Decreasing pressure
(C) Decreasing temperature Increasing pressure
(D) Decreasing temperature Decreasing pressure

Q12. Which of the following solution has zero pH?

(A) 0.5M HCl
(B) 0.5M HNO_3
(C) 0.5M H_2SO_4
(D) 0.5M CH_3COOH

Q13. For the reaction A → Product, if conc. of A is increased two times, the rate of reaction increases four time. The order of reaction is:

(A) 0
(B) 1
(C) 2
(D) 3

Q14. Which of the following is NOT colligative property?

(A) Lowering in vapor pressure
(B) Elevation in Boiling point
(C) Osmotic pressure
(D) Transition temperature

Q15. For the reactions involving solids and liquids, ΔH & ΔE are related as:

(A) $\Delta H = \Delta E$
(B) $\Delta H > \Delta E$
(C) $\Delta H < \Delta E$
(D) $\Delta H = 1/2\ \Delta E$

Q16. Which of following atoms in the given reaction undergoes reduction? $2Ag + H_2S + 1/2O_2 \rightarrow Ag_2S + H_2O$

(A) Ag
(B) S
(C) H
(D) O

Q17. Which of following metals cannot displace hydrogen from dil HCl?

(A) Mg
(B) Zn
(C) Fe
(D) Cu

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Q. SECTION – B (Marks 42)

Q. 2 Answer any FOURTEEN parts from the following. All parts carry equal marks. (14 x 3 = 42)

(i) What are the limiting and non-limiting reactants? Why expensive reactants are used in small amount whereas in-expensive reactants in large amount in an industrial process?

(ii) How much Fe is required to produce 200g of $FeCl_3$ when Cl_2 is present in excess? $2Fe + 3Cl_2 \rightarrow 2FeCl_3$ (Fe = 56 g/mol, Cl = 35.5 g/mol)

(iii) Why are x-rays produced when cathode rays collide with a target-anode? What are the types of spectral lines so produced?

(iv) Calculate the wave number of limiting line in the Balmer series of Hydrogen spectrum.

(v) Why the structure of $BeCl_2$ is different from that of $SnCl_2$? Explain by applying VSEPR theory

(vi) Describe any two factors which affect bond energy. Give one example for each.

(vii) What is absolute zero? Derive it from critical statement of Charle's law.

(viii) O_2 gas effuses 1.173 times as compared to an un-known gas. Calculate the molar mass of gas and identify it. The gas is produced during the combustion of hydrocarbons.

(ix) Why boiling points of noble gases increase from Helium to Radon?

(x) Justify the following order of boiling points of following hydrides. (a) H_2O > HF (b) HF > HCl (c) SiH_4 > CH_4

(xi) What is cleavage plane? Justify that cleavage plane is an anisotropic property.

(xii) Write down any three differences between metallic and molecular solids.

(xiii) How can a reaction at equilibrium be recognized by spectroscopic method?

(xiv) Describe the effect of increase in temperature on equilibrium position and equilibrium constant in the following reaction at equilibrium. $2SO_2 + O_2 \rightleftharpoons 2SO_3$ $\Delta H = -198kJ$

(xv) What is levelling effect? Why H_2O exhibits this phenomenon whereas CH_3COOH does not?

(xvi) pH of a buffer containing 1M CH_3COOH and 1M CH_3COONa is 4.76. Determine the change in pH on addition of 0.1M solution of NaOH to this buffer. pK_a = 4.76

(xvii) Rate equation for the reaction $2NO + 2H_2 \rightarrow N_2 + 2H_2O$ is Rate = $k[NO]^2[H_2]$. Suggest the mechanism for this reaction

(xviii) Describe the effect of temperature on solubility of solids and gases in liquids.

(xix) Determine the enthalpy of sublimation of I_2 from following data: $H_2(g) + I_2(g) \rightarrow 2HI(g)$ $\Delta H = +51.8kJ$; $H_2(g) + I_2(s) \rightarrow 2HI(s)$ $\Delta H = -10.5kJ$

(xx) What is Galvanizing? Why is it also called sacrificial corrosion?

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. Differentiate between Sp^3 and Sp^2 modes of hybridization. Explain the formation of Ethene molecule on the basis of hybridization in C-atoms. b. Give the reasons for the following facts: (i) Vapour pressure of Ether is higher than that of water. (ii) Food is cooked in less time in pressure cooker (iii) Vacuum distillation is carried out to purify sensitive liquids.

Q. 4 a. Why freezing point is decreased when a non-volatile, non-electrolyte solute is added to a solvent? Explain. Also predict the freezing point of a coolant prepared by dissolving 2kg Ethylene glycol in 5kg water (K_f = 1.86). b. State and explain Faraday's 1st and 2nd laws of electrolysis

Q. 5 a. Why real gas deviate from ideal behaviour? Explain these deviations at high pressure and low temperature with the help of diagram. b. Explain Hess's law, with an example. Write down its two indirect applications. How enthalpy of a reaction can be determined from heats of formation?

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VERSION SP

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Q1. The temperature at which partially immiscible pair of liquid leads to the formation of single phase is called:

(A) Absolute temperature
(B) Consolute temperature
(C) Standard temperature
(D) Transition temperature

Q2. In conversion of Br_2 to BrO_3^- the oxidation state of Bromine changes from:

(A) 0 to –3
(B) 0 to 5
(C) 2 to 5
(D) 0 to –2

Q3. In the given reaction, select the correct statement for the given reaction: $Mg + Cl_2 \rightarrow MgCl_2$

(A) Mg is reducing agent
(B) Cl_2 is reducing agent
(C) $MgCl_2$ is oxidizing agent
(D) Mg is oxidizing agent

Q4. The number of covalent bonds present in 2 moles of H_2S gas are:

(A) 6.02×10^{23}
(B) 2.4×10^{24}
(C) 24.1×10^{24}
(D) 3.01×10^{23}

Q5. The difference in angular momentum of electron which jumps from 3rd orbit to 6th orbit of Hydrogen atom will be:

(A) $3(h/\pi)$
(B) $6(h/\pi)$
(C) $6(h/\pi^2)$
(D) $3(h/2\pi)$

Q6. Which of the following compounds has the highest ionic character?

(A) HCl
(B) HBr
(C) HI
(D) HF

Q7. A diatomic molecule has dipole moment of 1.2D. If its bond distance is 1.0A°. What fraction of electric charge exists in each atom?

(A) 18% of e^-
(B) 25% of e^-
(C) 30% of e^-
(D) 12% of e^-

Q8. Gases exert pressure on the walls of the container, it is due to the:

(A) Elastic collision between their molecules
(B) Vander Wall's forces between their molecules
(C) Force of repulsion between their molecules
(D) Force of attraction between their molecules

Q9. A liquid is thought to be pure Diethyl ether. Which of the following is the best way to test its purity?

(A) Burn with oxygen
(B) Dehydrate with H_2SO_4
(C) React with pure ether
(D) Measure its boiling point

Q10. Amount of heat absorbed when one mole of a solid melts into liquid form at its melting point is called:

(A) Molar heat of vaporization
(B) Molar heat of condensation
(C) Molar heat of fusion
(D) Molar heat of sublimation

Q11. NaF and KCl have both atomic ratio 1:1 in their crystals, such property is called:

(A) Isomorphism
(B) Allotropy
(C) An isotropy
(D) Polymorphism

Q12. Diamond is a bad conductor of electricity because:

(A) It has high density
(B) It has no free electrons
(C) It is transparent to light
(D) It has tight structure

Q13. Identify what is TRUE about the following reaction: $H_2 + Cl_2 \rightleftharpoons 2HCl$

(A) $K_p < K_c$
(B) $K_p = K_n$
(C) $K_n < K_c$
(D) $K_p > K_c$

Q14. Which of the following change will favour the formation of more SO_3 at equilibrium? $2SO_2 + O_2 \rightleftharpoons 2SO_3 + heat$

(A) By increasing temperature
(B) By decreasing pressure
(C) By increasing pressure
(D) By adding SO_3 at equilibrium

Q15. The concentration of H^+ ion concentration of an aqueous solution having pH = 10.6 is:

(A) $1.02\times10^{-11} mol.dm^{-3}$
(B) $1.07\times10^{-2} mol.dm^{-3}$
(C) $2.15\times10^{-9} mol.dm^{-3}$
(D) $2.51\times10^{-11} mol.dm^{-3}$

Q16. Rate law for the reaction $CH_3–Cl + H_2O \rightarrow CH_3OH + HCl$, Rate = $K[CH_3–Cl]$. The rate of reaction will be doubled when:

(A) Concentration of CH_3Cl is reduced to half
(B) Concentration of CH_3Cl is doubled
(C) Concentrations of CH_3Cl and H_2O are reduced to half
(D) Concentration of H_2O is doubled

Q17. Molarity of acetic acid (CH_3COOH) solution is ____ when 9g of it is dissolved in $0.25 dm^3$ of solution.

(A) 0.6M
(B) 0.3M
(C) 0.5M
(D) 0.25M

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Q. SECTION – B (Marks 42)

Q. 2 Answer any FOURTEEN parts from the following. All parts carry equal marks. (14 x 3 = 42)

(i) What is meant by percentage composition? Determine the mass percentage of each element in magnesium chloride ($MgCl_2$).

(ii) (a) What is the relationship between wave number and frequency? (b) What will be the wave number ($\bar{\nu}$) of the spectral lines of an electron when it jumps from $n_1 = 2$ to $n_2 = 4$?

(iii) Which subshell in each of the following pairs is higher in the energy? (a) 4p and 4s (b) 5s and 6p (c) 6s and 4f

(iv) How Moseley used x-rays spectrum to predict the atomic number of elements. Give one use of x-rays in the field of chemistry.

(v) The molecules of $AlCl_3$ have planar triangular geometries while NH_3 have triangular pyramidal geometries. Briefly explain on the basis of VSEPR theory.

(vi) State Joule-Thomson effect. Give its one application in home appliances.

(vii) Calculate average molar mass of air at sea level and 0°C if the density of air is 1.29 kg/m^3

(viii) Give the relationship of partial pressure with: (a) Number of moles of gas (b) Mole fraction of gas

(ix) Why distillation under reduced pressure is often used in the purification of chemicals? Elaborate with an example of glycerine.

(x) Describe the conductivity of metallic crystals using electron sea theory.

(xi) Briefly explain the effect of change in pressure on the volume of a gas, when temperature remains constant. Give graphical representation.

(xii) Describe the following properties of crystalline solids. (a) Transition temperature (b) Anisotropy (c) Polymorphism

(xiii) How would the solution of CH_3COOH and CH_3COONa resist the change in pH upon addition of HCl?

(xiv) Prove the given relationship pK_a + pK_b = pK_w by selecting general reaction of dissociation for acid and base.

(xv) The rise in temperature increases the rate of a certain chemical reaction. Briefly explain in the light of Collision Theory by graphical representation.

(xvi) Differentiate between the First order reaction and Pseudo first order reaction with suitable example in each case.

(xvii) Briefly explain heat of solution. Apply this concept to the hydration of ammonium nitrate crystals. $\Delta H_sol = 2.57\ KJmol^{-1}$

(xviii) Calculate ΔH° for the given reaction $NO + O \rightarrow NO_2$ from the following data. $NO + O_3 \rightarrow NO_2 + O_2$ ΔH° = –198.9 KJ; $O_3 \rightarrow 3/2\ O_2$ ΔH° = –142.3 KJ; $O_2 \rightarrow 2O$ ΔH° = 495.0 KJ

(xix) The standard reduction potentials for the following half reactions are: $Ni^{+2} + 2e^- \rightarrow Ni(s)$ $E° = -0.25V$; $Mg^{+2} + 2e^- \rightarrow Mg(s)$ $E° = -2.38V$. Calculate E°cell. Write cell reaction. Show the direction of electrons.

(xx) Describe how a dry cell produces electricity and helps to operate the electric appliances.

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. 30g of K_2SO_4 on dissolving in water ionizes completely. Calculate: (i) Number of K_2SO_4 molecules (ii) Number of K^+ and SO_4^{-2} ions (iii) Mass of individual ions. b. Why is Molecular Orbital Theory superior to Valence Bond Theory. Explain the magnetic behaviour of N_2, N_2^{+1} and N_2^{-2} species on the basis of Molecular Orbital Theory.

Q. 4 a. What is hydrogen bonding? Using the concept of Hydrogen bonding in water, explain the given properties: (i) High surface tension (ii) High specific heat (iii) High heat of vaporization. b. Design best three conditions to get maximum yield of ammonia by using given chemical reaction. $N_2 + 3H_2 \rightleftharpoons 2NH_3$ $\Delta H° = -92.46KJ$

Q. 5 a. What is cryoscopic constant? Discuss the quantitative aspects of depression in freezing point of a solution. Give graphical representation for depression in freezing point. b. Born Haber's cycle is special application of Hess's law for Binary ionic compound. Write reactions and draw step wise Born Haber cycle for measurement of $\Delta H°$ lattice for sodium chloride (NaCl).

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